Yakeen 2.0 2027

Replica Practice Sheet

Physics by MR Sir

Basic Maths and Calculus

Q1 Find value of $\frac{10^{-4}}{8}$

(A) 1.25×10^{-5}

(B) 1.25×10^{-4}

(C) 1.25×10^{-3}

(D) 1.25×10^{-6}

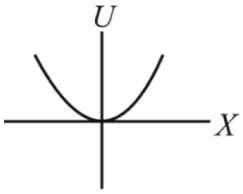
Q2 If $y = \sin 2\theta$ then find 'θ' where y will be maximum:

(A) 90° (B) 60°

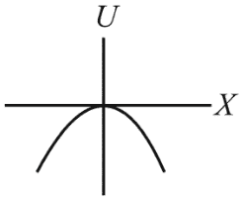
(C) 45° (D) 32°

Q3 A body is attached to a spring whose other end is fixed. If the spring is elongated by x, its potential energy is $U = 5x^2$, where x is in metre and U is in joule. $U - x$ graph is

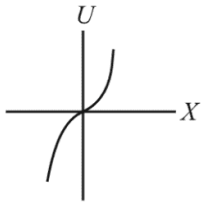
(A)



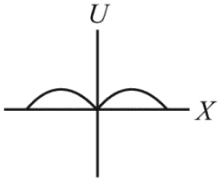
(B)



(C)



(D)



Q4 **Statement I**- As θ increases, the value of $\cos\ \theta$ also increases $(0^\circ \leq \theta \leq 90^\circ)$

Statement II - For a very small angle θ, $\sin\theta \simeq \theta$

(A) Statement I is correct but Statement II is incorrect.

(B) Statement I is incorrect but Statement II is correct.

(C) Both Statement I and Statement II are correct.

(D) Both Statement I and Statement II are incorrect.

Q5 Choose the **correct** statement(s) among the following.

(I) The integral $\int_1^5 x^2\ dx$ is equal to $\frac{124}{3}$

(II) The value of $(\sin\ 180^\circ\ +\ \cos\ 90^\circ)^2$ is 1

(III) Slope of straight line $\frac{x}{2} - \frac{y}{4} = 1$ is 2.

(A) Only I

(B) I and II

(C) I, II and III

(D) I and III

Q6 The slope of straight line $\sqrt{3}y = 3x + 4$ is:

(A) 3 (B) $\sqrt{3}$

(C) $\frac{1}{\sqrt{3}}$ (D) $\frac{1}{3}$

Q7 $\int_0^1 e^x dx$

(A) e^x (B) 1

(C) 0 (D) $e - 1$

Q8 If $B_{\text{axis}} = B_{\text{centre}} \left(\frac{R^3}{(R^2+x^2)^{3/2}}\right)$, then find $\frac{B_{\text{axis}}}{B_{\text{centre}}}$ if $x << R$

(A) $\left[1 - \frac{3}{2}\frac{x^2}{R^2}\right]$ (B) $\left[1 + \frac{3}{2}\frac{x^2}{R^2}\right]$

(C) $\left[1 + \frac{3}{2}\frac{x}{R}\right]$ (D) $\left[1 - \frac{3}{2}\frac{x}{R}\right]$

Q9 The equation of a curve is given as $y = x^2 + 2 - 3x$. The curve intersects the y-axis at:

(A) (0, 1) (B) (2, 0)

(C) (0, 2) (D) (1, 0)

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Q10 If $\cos\theta = \frac{4}{5}$ then find the value of $\tan\theta$

(A) $\frac{4}{5}$ (B) $\frac{3}{5}$

(C) $\frac{4}{3}$ (D) $\frac{3}{4}$

Q11 Sound intensity level (in decibel) is given by the formula, $\beta = 10 \log \frac{I}{I_o}$. Here given that $I = 2 \times 10^2 \ W/m^2$ and $I_o = 2 \times 10^{-12} \ W/m^2$. Calculate β

(A) 10 (B) 140

(C) 14 (D) 130

Q12 Find the value of $(64)^{2/3}$

(A) 4 (B) 16

(C) 32 (D) 64

Q13 If x+y = 8, then what will be the maximum value of xy ?

(A) 8 (B) 16

(C) 20 (D) 24

Q14 **Assertion :** Distance between two points (1, 2, 3) and (1,6,6) is 5 units.

Reason : The distance between two points (x_1, y_1, z_1) and (x_2, y_2, z_2) is given by the formula : $r = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2 + (z_2-z_1)^2}$

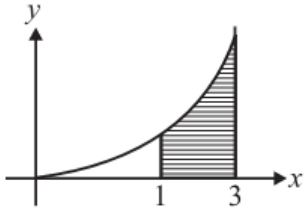
(A) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.

(B) Both Assertion and Reason are correct, but Reason is not the correct explanation of Assertion.

(C) Assertion is correct, but Reason is incorrect.

(D) Assertion is incorrect, but Reason is correct.

Q15 Find the area under the shaded region for curve $y = 3x^2$.



(A) 80 (B) 26

(C) 20 (D) 40

Q16 Find $1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \frac{1}{16} - \frac{1}{32} + ...\infty$

(A) 2 (B) 1

(C) 2/3 (D) ∞

Q17 $\log 25 + \log 4 - \log 5$ is equal to

(A) $\log 20$ (B) $\log 25$

(C) $\log 15$ (D) $\log 10$

Q18 $\frac{d}{dx}\left(e^{100}\right) = ...$

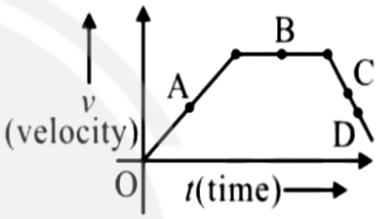
(A) e^{100}

(B) 0

(C) 100

(D) 1

Q19 The slope of $v - t$ is zero at point



(A) A (B) B

(C) C (D) D

Q20 Evaluate $\int \frac{dx}{\sqrt[3]{x}}$

(A) $\frac{3}{2}x^{-2/3} + c$

(B) $\frac{2}{3}x^{-3/2} + c$

(C) $\frac{2}{3}x^{3/2} + c$

(D) $\frac{3}{2}x^{2/3} + c$

Q21 If $\frac{\sin\theta+\cos\theta}{\sin\theta-\cos\theta} = \frac{7}{3}$ then find $\tan\theta$?

(A) $\frac{3}{5}$ (B) $\frac{5}{2}$

(C) $\frac{5}{3}$ (D) $\frac{2}{5}$

Q22 Calculate $\frac{9/8}{6/5}$

(A) $\frac{16}{15}$ (B) $\frac{15}{16}$

(C) $\frac{5}{16}$ (D) $\frac{27}{20}$

Q23 Find the approximate value of $(1+x)^3$, if $|x| << 1$.

(A) $1+x$

(B) $1-3x$

(C) $1+3x$

(D) $1-3x+3x^2+x^3$

Q24 Find the value of $\log_{10} 1000 - \log_{10} 100$ = ____?

(A) 3 (B) 2

(C) 1 (D) 10

Q25 If $y = \log_e x + \sin x + e^x$, then $\frac{dy}{dx}$ is:

(A) $\frac{1}{x} + \sin x + e^x$

(B) $\frac{1}{x} - \cos x + e^x$

(C) $\frac{1}{x} + \cos x + e^x$

(D) $\frac{1}{x} - \sin x$

Q26 Match the List I with List II to find out the **correct** option

	List I		**List II**
(i).	$\log_e 125 + \log_e 4 - 2\log_e 5$	(a)	1
(ii).	$\log_e 16$	(b)	$\log_e 20$
(iii).	$\log_{10} 10$	(c)	$4 \log_e 2$
(iv).	$\log_2 16$	(d)	4

Choose the **correct** option from the codes given below

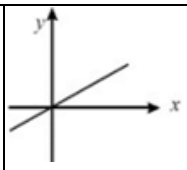
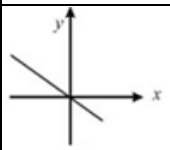
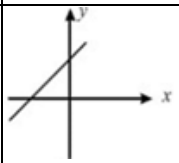
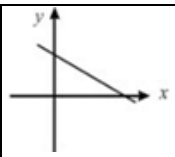
(A) i-(b), ii-(a), iii-(c), iv-(d)

(B) i-(a), ii-(b), iii-(c), iv-(d)

(C) i-(b), ii-(c), iii-(a), iv-(d)

(D) i-(d), ii-(a), iii-(b), iv-(c)

Q27 Match the graph in **List-II** corresponding to the equations given in **List-I.**

List I		**List II**	
(i).	$y = 4x$	(a)	
(ii).	$y = -6x$	(b)	
(iii).	$y = x + 4$	(c)	
(iv).	$y = -2x + 4$	(d)	

Choose the **correct** option from the codes given below

(A) i-(b), ii-(c), iii-(d), iv-(a)

(B) i-(a), ii-(d), iii-(b), iv-(c)

(C) i-(b), ii-(c), iii-(a), iv-(d)

(D) i-(a), ii-(b), iii-(c), iv-(d)

Q28 Convert the following angles from radian to degree

(a) $\frac{3\pi}{4}$rad

(b) $\frac{7\pi}{6}$rad

(A) 135°, 210°

(B) 210°, 135°

(C) 225°, 240°

(D) 135°, 225°

Q29 If $y = \frac{1}{2}\sin(x^2)$, $\frac{dy}{dx}$ will be:

(A) $\frac{1}{2}\cos(x^2)$

(B) $x\cos(x^2)$

(C) $\cos(x^2)$

(D) $\sin(x)$

Q30 Find the values of:

(i) tan (– 30°)

(ii) cos 150°

(iii) sin 210°

(A) $\frac{1}{\sqrt{3}}, \frac{\sqrt{3}}{2}, \frac{1}{2}$ (B) $-\frac{1}{\sqrt{3}}, -\frac{\sqrt{3}}{2}, -\frac{1}{2}$

(C) $-\frac{1}{\sqrt{3}}, \frac{\sqrt{3}}{2}, \frac{1}{2}$ (D) $-\frac{1}{\sqrt{3}}, \frac{\sqrt{3}}{2}, -\frac{1}{2}$

Q31 If acceleration due to gravity g at height $h \ll R$ (where R is radius of earth) is $g_h = g_0\left(1 + \frac{h}{R}\right)^{-2}$, then using binomial theorem which is **correct** ?

(A) $g_h = g_0$

(B) $g_h = g_0\left(1 - \frac{2h}{R}\right)$

(C) $g_h = g_0\left(1 + \frac{2h}{R}\right)$

(D) $g_h = g_0\left(1 - \frac{h}{2R}\right)$

Q32 $\sqrt{1 + \sin\theta}$ is equal to

(A) $|\sin\theta + \cos\theta|$

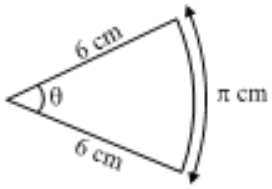
(B) $|\sin\theta - \cos\theta|$
(C) $\left|\sin\frac{\theta}{2} + \cos\frac{\theta}{2}\right|$
(D) $\left|\sin\frac{\theta}{2} - \cos\frac{\theta}{2}\right|$

Q33 If $y^2 - 2y - 3 = 0$, then find the value of y.
(A) 3, 1
(B) –3, –1
(C) 3, –1
(D) –3, 1

Q34 Find approximate value of: $(1.005)^{12}$
(A) 1.005 (B) 1.060
(C) 1.025 (D) 1.020

Q35 The radius r of spherical bubble is changing with time t. The rate of change of its volume is given by:
(A) $4\pi r^2 \frac{dr}{dt}$
(B) $\frac{4}{3}\pi r^2$
(C) $\frac{8}{3}\pi r^2$
(D) $\frac{8}{3}\pi r \frac{dr}{dt}$

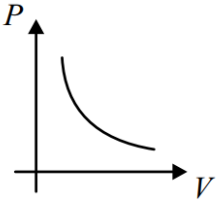
Q36 A circular arc is of length π cm. Find angle subtended by it at the centre.



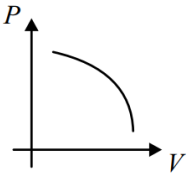
(A) 60° (B) 30°
(C) 90° (D) 15°

Q37 Find the value of integral $\int_0^{\pi/2} \cos\, x\, dx$
(A) 0 (B) 1
(C) -1 (D) 2

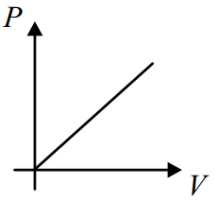
Q38 P-V graph for ideal gas at constant temperature(T) is [Given ideal gas equation PV =nRT]
(A)



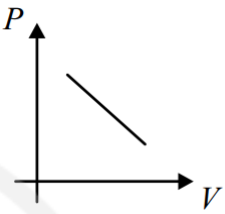
(B)



(C)



(D)



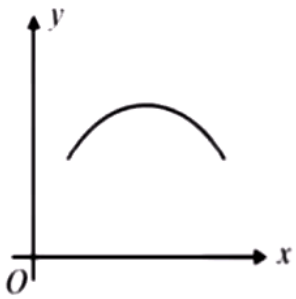
Q39 The minimum value of $y = 5x^2 - 2x + 1$ is
(A) 1/5 (B) 2/5
(C) 4/5 (D) 3/5

Q40 $y = 2u^3, u = 8x - 1$. Find $\frac{dy}{dx}$
(A) $48(8x-1)^2$
(B) $48(8x+1)^2$
(C) $48(8x-1)$
(D) $48(8x+1)$

Q41 The area of the region between the given curve $y = 4x^3$ and the x–axis on the interval [0, 1] is:
(A) 2 unit (B) 0
(C) 3 unit (D) 1 unit

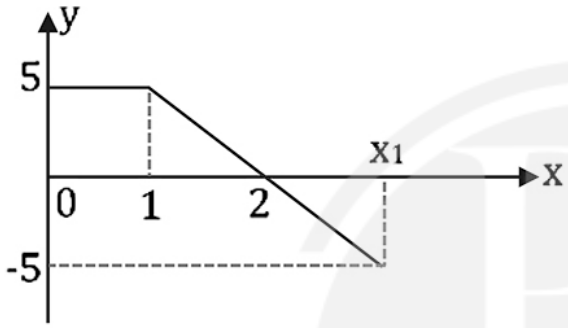
Q42 Evaluate $\int \left(x^2 - \cos x + \frac{1}{x}\right)dx$
(A) $x^3 - \sin\ x + \ln(x) + c$
(B) $2x - \sin\ x + \ln(x) + c$
(C) $\frac{x^3}{3} + \sin x + \ln(x) + c$
(D) $\frac{x^3}{3} - \sin x + \ln(x) + c$

Q43 Magnitude of slope i.e., steepness of graph shown in figure



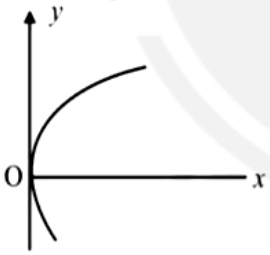
(A) First increase and then decreases
(B) First decreases and then increases
(C) Decreases continuously
(D) Increases continuously

Q44 Find the value of x_1, so that $\int_0^{x_1} y dx = 5$



(A) 2
(B) 7
(C) 3
(D) 5

Q45 At $x = 0$, the value of slope is



(A) 0
(B) 1
(C) –1
(D) Infinite

Answer Key

Q1	(A)
Q2	(C)
Q3	(A)
Q4	(B)
Q5	(D)
Q6	(B)
Q7	(D)
Q8	(A)
Q9	(C)
Q10	(D)
Q11	(B)
Q12	(B)
Q13	(B)
Q14	(A)
Q15	(B)
Q16	(C)
Q17	(A)
Q18	(B)
Q19	(B)
Q20	(D)
Q21	(B)
Q22	(B)
Q23	(C)
Q24	(C)
Q25	(C)
Q26	(C)
Q27	(D)
Q28	(A)
Q29	(B)
Q30	(B)
Q31	(B)
Q32	(C)
Q33	(C)
Q34	(B)
Q35	(A)
Q36	(B)
Q37	(B)
Q38	(A)
Q39	(C)
Q40	(A)
Q41	(D)
Q42	(D)
Q43	(B)
Q44	(C)
Q45	(D)


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